

Defining the variable

Model: package masses are approximately normal with mean 500 g and standard deviation 20 g.

What does X represent if we select one package?

A lower-tail probability

Area to the left of a cutoff gives a lower-tail probability. Model: $X \sim N(500, 20)$, in grams.

What is the probability that one package weighs less than 480 g?

An upper-tail probability

Area to the right of a cutoff gives an upper-tail probability. Model: $X \sim N(500, 20)$, in grams.

What is the probability that one package weighs more than 530 g?

A probability between cutoffs

Model: $X \sim N(500, 20)$, measured in grams.

What is $P(490 < X < 520)$?

Outside an acceptable range

Model: package masses follow $N(500, 20)$. The accepted range is 470 to 530 g.

What proportion falls outside the accepted range?

Strict or inclusive?

Model: exact package mass has a continuous normal distribution.

Does $P(X < 480)$ differ from $P(X \leq 480)$?

A technology input

A normal cumulative probability tool needs a mean, standard deviation, and cutoff or interval.

For $P(X > 530)$ when $X \sim N(500, 20)$, what information should you enter?

Shading a normal probability

Model: package mass $X \sim N(500, 20)$, in grams.

How would you sketch and shade the region for $P(490 < X < 520)$?

Checking the direction

Model: $X \sim N(500, 20)$. A calculation for $P(X > 530)$ returns 0.9332.

Is that result plausible for a cutoff above the mean?

From probability to expected count

Model: 8% of packages fall outside specification. A shipment contains 250 packages.

How many out-of-specification packages should we expect?

A percentile

A percentile cutoff specifies area at or below that value.

What probability equation defines the 90th percentile k ?

Finding a percentile

Model: $X \sim N(500, 20)$, measured in grams.

What mass is the 90th percentile?

Inverse normal input

An inverse normal calculation returns the cutoff for a specified area to its left. Model: $X \sim N(500, 20)$.

What inputs find the mass exceeded by only 10% of packages?

An upper-tail cutoff

Model: $X \sim N(500, 20)$, measured in grams.

Only 5% of packages exceed what mass?

Critical value

A critical value is a cutoff marking a specified probability region. Such cutoffs will be used in confidence intervals and tests.

What standard-normal critical value leaves area 0.025 to its right?

A lower-tail cutoff

Model: $X \sim N(500, 20)$, measured in grams.

What mass separates the lowest 10% from the rest?

The middle 90%

Model: $X \sim N(500, 20)$, measured in grams.

What symmetric interval around the mean contains the middle 90%?

Quartiles from a model

Model: $X \sim N(500, 20)$, measured in grams.

What masses are the first and third quartiles?

An interquartile range

Model: $X \sim N(500, 20)$, measured in grams.

What is the model's interquartile range?

Different spreads

Model A is $N(500, 10)$. Model B is $N(500, 20)$, with mass in grams.

Which model gives a greater probability of mass above 530 g?

Different centers

Model A is $N(500, 20)$. Model B is $N(510, 20)$, with mass in grams.

Which model gives a greater probability of mass below 480 g?

A probability or a cutoff?

A manager asks for the mass exceeded by only 2% of packages.

Does the requested answer have units of grams or units of probability?

Conditional normal probability

Model: $X \sim N(500, 20)$. A package is known to weigh more than 500 g.

What is the probability that it weighs more than 520 g?

A model boundary

A normal model assigns some probability to every real-valued interval.

When could its prediction of negative durations make it unsuitable for duration data?

One package or an average?

Model: independent package masses follow $N(500, 20)$.

Should an average of 100 masses be analyzed using the same standard deviation as one mass?

Checking an empirical normal model

Synthetic durations (s): 8, 9, 9, 10, 10, 10, 11, 11, 12, 20. Fit a normal model using the sample mean and SD.

How does its predicted proportion above 12 compare with the observed proportion?